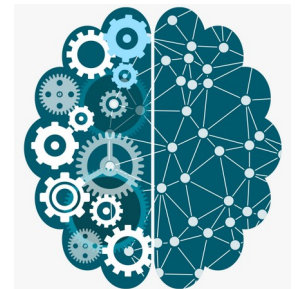


Mastering Python OS module

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What is the os Module?

- Provides a portable way of using operating system-dependent functionality.
- Works cross-platform (Windows, macOS, Linux).
- Allows interaction with the file system, directories, and environment variables.
- Part of the Python Standard Library (no installation required).
- How to import?
 - `import os`

Directory Operations: Where Are We?

- `os.getcwd()`: Get Current Working Directory. Returns the absolute path where your script is executing.
- `os.chdir()`: Changes the current working directory.

```
# Get current directory
print(os.getcwd())
# Move up one directory
os.chdir('..')
```

Directory Operations: Creating & Deleting

- `os.mkdir()`:
 - Creates a single new directory.
- `os.makedirs()`:
 - Creates nested directories (creates intermediate folders if they don't exist).
- `os.rmdir()`:
 - Removes a single empty directory.
- `os.removedirs()`:
 - Removes nested empty directories.

Directory Operations: Creating & Deleting

- `os.makedirs('parent/child', exist_ok=True)`
- `os.rmdir('my_empty_folder')`

Listing and Modifying Files

- `os.listdir()`: Returns a list of all files and folders in a specific directory.
- `os.rename()`: Renames a file or folder.
- `os.remove()`: Deletes a specific file.

```
# List contents
print(os.listdir('.'))
# Rename and delete
os.rename('old.txt', 'new.txt')
os.remove('new.txt')
```

Getting File Details with os.stat()

- Used to extract metadata about a file or directory.
- Can retrieve file size (st_size).
- Can retrieve modification time (st_mtime), creation time, and more.

```
import time
stats = os.stat('data.txt')
print(f"Size: {stats.st_size} bytes")
print(f"Modified: {time.ctime(stats.st_mtime)}")
```

The os.path Submodule (Part 1)

- Crucial for cross-platform compatibility (handles \ vs / automatically).
- `os.path.join()`: Safely combines folder and file names.
- `os.path.exists()`: Checks if a specific path exists on the system.

```
# Safely join paths
full_path = os.path.join('docs', 'report.pdf')

# Check existence
if os.path.exists(full_path):
    print("File found!")
```


The os.path Submodule (Part 2)

- Identify path types: `os.path.isfile()` and `os.path.isdir()`.
- Extract components: `os.path.basename()` (file name) and `os.path.dirname()` (folder path).
- Split extensions: `os.path.splitext()`.

```
path = "/users/admin/budget.xlsx"  
print(os.path.basename(path)) # budget.xlsx
```

```
name, ext = os.path.splitext('budget.xlsx')  
# name: 'budget', ext: '.xlsx'
```

Traversing a Directory Tree

- `os.walk()` generates the file names in a directory tree by walking it either top-down or bottom-up.
- Yields a 3-tuple: (dirpath, dirnames, filenames).
- Perfect for bulk file processing or searching.

```
for dirpath, dirs, files in os.walk('my_folder'):  
    print(f"Current Dir: {dirpath}")  
    for file in files:  
        print(f"Found: {file}")
```

Managing Environment Variables

- Useful for handling sensitive data (API keys, configurations) securely outside of source code.
- `os.getenv()`: Safely retrieves a variable (returns `None` if not found, or a specified default).
- `os.environ`: A dictionary mapping representing the system's environment variables.

```
# Get a variable safely  
api_key = os.getenv('API_KEY', 'default_value')
```

```
# Set a variable (for current session only)  
os.environ['MY_VAR'] = 'Secret123'
```

Modifying Permissions with `os.chmod()`

- Changes the access permissions of a file (Read, Write, Execute).
- Most effective on Unix-like systems (Linux/macOS).
- Used in combination with the built-in `stat` module for readable constants.

```
import stat

# Make a file read-only for the owner
os.chmod('secret.txt', stat.S_IRUSR)

# Add Read + Write + Execute
os.chmod('script.sh', stat.S_IRUSR |
stat.S_IWUSR | stat.S_IXUSR)
```

Summary

- os provides fundamental tools for interacting with your operating system.
- Always use `os.path.join` for file paths to avoid cross-platform bugs.
- Use `os.walk` for deep directory searches.
- Keep secrets safe using `os.getenv()`.

Thank you

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